Department of Computer Science & Engineering (CSE)
School of Engineering, Technology & Sciences

INDEPENDENT UNIVERSITY, BANGLADESH (IUB)

Computer Science and Engineering (CSE) Program (*Undergraduate*)

COURSE OUTLINE	**Spring 2025**	**FULL MARKS: 100**

Course Code	CSE200	**Course Title**	Computer Programming II
Course Type	Core	**Prerequisite (if any)**	CSE101 + Lab
Credit Value	4 (T: 3, L: 1)	**Contact Hours / Week**	3 hours (theory) 2 hours (Lab)
Year	1st Year	**Semester**	2nd Semester

COURSE DESCRIPTION

The main objective of this course is to develop a strong foundation of computer programming knowledge within the CSE students for real-life problem solving from the perspective of a software engineer. Students will gain mastery of the concepts of both procedural language and object-oriented (OOP) paradigm. The advance knowledge of control statements (e.g., conditional statements, loops), program modularization (e.g., functions, class and methods), data storage, management, and manipulation (e.g., arrays, list, sets, tuple, dictionary), and secondary storage management (e.g., File system) are covered in detail. The concept of object-oriented programming, e.g., class, object, methods, inheritance, polymorphism, encapsulation are introduced and used to solve problems. Additionally, error handling mechanisms (e.g., exception handling) are also discussed. Each lecture would involve solving several programming problems on the learned topic using designated IDE. After successful completion of the course, a student will possess the mastery of breaking down a complex programming problem into smaller parts, solve them and write the solution in python.

COURSE OBJECTIVE

a. Advance skill on the understanding and use of structured programming concepts in solving real-life computational problems as sketched above.
b. Detailed understanding of the OOP paradigm and developing skills to use them in solving real-life computational problems.
c. Learn in detail the memory management (both primary and secondary), data structure and error handling mechanism.
d. Grow in confidence with programming skills in dealing with non-trivial problem solving.

COURSE POLICY

It is the student's responsibility to gather information about the assignments and cover topics if they miss lectures. Regular class attendance is mandatory. Following policy as per the faculty manual will be adopted: Points will be taken off for missing class tests. Without 70% of attendance, sitting for the final exam is NOT allowed. According to the IUB policy, students must enter the classroom within the first 10 minutes to get the attendance submitted.

a. The date and syllabus of the lectures, quizzes, midterm, and final exam are already given here, however, announcements will be given ahead of time. There is **NO** provision for make-up quizzes.

b. The reading materials for each class may be given before that class so that students may have a cursory look into the materials. All materials (lecture notes, supporting reading materials, problem sheets and the solutions) will be made available through google classroom (classroom.google.com).
c. Class participation is vital for a better understanding of the subject matter. Class will be conducted in an interactive environment where both the teacher and the students must pose questions and discuss solutions for better understanding.
d. Students must attend the scheduled tutorial classes of the Research Assistants. Attending and performing in the tutorial session is mandatory and is part of the overall course evaluations.
e. Students must maintain the IUB code of conduct and ethical guidelines offered by the school of engineering and computer sciences.

ACADEMIC DISHONESTY

a. A student who cheats, plagiarizes, or furnishes false, misleading information in the course is subject to disciplinary action up to and including an F grade in the course and/or suspension/expulsion from the University.
b. Students must maintain the code of IUB.
c. The goal of homework is to give you practice in mastering the course material. Consequently, you are encouraged to collaborate on problem sets. In fact, students who form study groups generally do better on exams than do students who work alone. If you do work in a study group, however, you owe it to yourself and your group to be prepared for your study group meeting. Specifically, you should spend at least 30-45 minutes trying to solve each problem beforehand by yourself. If your group is unable to solve a problem, talk to other groups or ask your recitation instructor or teaching assistant assigned to your class.
d. You must write up each problem solution by yourself without assistance. It is a violation of this policy to submit a problem solution that you cannot orally explain to a member of the course staff.
e. No collaboration whatsoever is permitted during examination.
f. Plagiarism and other anti-intellectual behavior cannot be tolerated in any academic environment that prides itself on individual accomplishment. If you have any questions about the collaboration policy, or if you feel that you may have violated the policy, please talk to one of the course staff. Although the course staff is obligated to deal with cheating appropriately, we are more understanding and lenient if we find out from the transgressor himself or herself rather than from a third party or by ourselves.

STUDENT WITH DISABILITIES AND STRESS

Students with disabilities are required to inform the Department of Computer Science & Engineering of any specific requirement for classes or examination as soon as possible. Additionally, if you experience significant stress or worry, changes in mood, or problems eating or sleeping this semester, whether because of this or any other courses or factors, please do not hesitate to reach out immediately, at any hour, to any of the course's heads to discuss.

NON DISCRIMINATION POLICY

The course and University policy prohibit discrimination based on race, color, religion, sex, marital or parental status, national origin or ancestry, age, mental or physical disability, sexual orientation, military status. If you see either by the course instructor or any other person related to the course showing any form of discrimination, please inform the proctor's office of the wrongdoing.

COURSE CONTENT

a. Conditional Statements (Nested) **1 Week (3h T + 1.5h L)**

Use of conditional statements in solving complex decision-making problems. Learn the use of nested conditional statements and associated technical details: if-else, if-elif-else, nested if.

b. Loops (Nested) **1 Week (3h T + 1.5h L)**

Solving complex recurring problems using loops. Learn in detail: while loop, for loop, nested loop, break statement, continue statement, pass statement.

c. Functions – calls, parameters, scope of variables **1 Week (3h T + 1.5h L)**

Introduce the concept of functions with function calls, use of parameters while calling functions and realization of the variables scope in the context of functions

d. Recursion **1 Week (3h T + 1.5h L)**

Detailed understanding of the recursive function calls with its behavior and realizing it with a physical world example.

e. Data structure (List, Tuple, Set, Dictionary) **3 Weeks (9h T + 4.5h L)**

Detailed understanding on the data storage and management and its importance from real life problem scenarios. Define, store, and manipulate List, Tuple, Set and Dictionary for data management. Learn in detail associated library methods and techniques. Multidimensional data structures, e.g., 2D list should be learnt.

f. Modularization (Function) and Introduction of a Library **2 Weeks (6h T + 3h L)**

Define function from the perspective of modularization / reusability / maintainability / separation of concept / divide-and-conquer in solving complex and recurring problems, Use of function: Argue from real life usage scenario (e.g., an automated chocolate factory), Show difference between code solution with and without function, Allow student to solve problems (conceptually) using functions, Structure of a function, e.g., parameters, returns value, calling a function, return from a function, With simple example show step-by-step how a function executes. Let the student understand how a program execution flows when a function is called, executed, and returned, Example problems and their implementation using functions.

g. Files (File I/O) **1 Week (3h T + 1.5h L)**

Introduce the concept of File as a secondary storage structure / persistence storage, what is the distinction between in memory storage (e.g., variables / data structures) and File, learn how to create and manipulate (read/write/append) file, Read / write data back and forth in a file. Design and implement a real-life problem using files.

h. Exception Handling **1 Week (3h T + 1.5h L)**

Introduce to the concept of exception and its distinction with conventional error. How to handle exceptions - exception handling and assertions.

i. Python - Class, Object **1 Week (3h T + 1.5h L)**

The concept of Class, Objects, attributes (class variables and instance variables), methods, accessing methods and attributes.

j. Review **2 Weeks (6h T + 3h L)**

COURSE LEARNING OUTCOME (CLO) MATRIX								
CLOs	**CO Description**	Blooms Learning Level*				PLO Assessed	CLO – PLO Correlation **
		C	P	A	S		
CLO1	Analyze an engineering programming problem to formulate a comprehensive solution using a specified programming language.	3				PLO-b	3
CLO2	Determine and apply the knowledge of advanced programming logic to solve simple real-life computational problems.		3			PLO-c	3
CLO3	Develop a comprehensive solution to a real-life programming problem with justification in relation to design alternatives, and selection of specific programming principles.		3			PLO-c	3

***Bloom's Learning Level:** Numbers signifies the Level of Bloom's skills.

****CLO – PLO Correlation:** 3 – high, 2 – medium, 1- low

PLO-b: Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences (K1 to K4).

PLO-c: Design/Development of solutions: Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (K5)

LESSON PLANNING WITH MAPPING OF CLO, TEACHING AND ASSESSMENT STRATEGIES

Week	Topic	Teaching-Learning Strategy	Assessment Strategy	Corresponding CLOs
1	Conditional Statements (Nested)	Lecture (3h) Lab (1.5h).	Class Test/Mid term	
2	Loops (Nested)	Lecture (3h), Lab (1.5h).	Class Test / Mid Term / Lab work Progressive project	CLO1
3	Functions - calls, parameters, scope of variables, generators	Lecture (3h), Lab (1.5h).	Class Test / Mid Term / Lab work Progressive project	CLO1 CLO2
4	Recursion	Lecture (3h), Lab (1.5h).	Class Test / Mid Term / Lab work Progressive project	CLO1 CLO2
5	Built-in data types - List, Tuple, String, Set, Dictionary	Lecture (3h), Lab (1.5h).	Class Test / Mid Term / Lab work Progressive project	CLO1 CLO2
6	Built-in types - List, Tuple, Set, Dictionary	Lecture (3h), Lab (1.5h).	Class Test / Mid Term / Lab work Progressive project	CLO1 CLO2

7	Built-in data types - List, Tuple, Set, Dictionary	Lecture (3h), Lab (1.5h).	Class Test / Mid Term / Lab work Progressive project	CLO1 CLO2
Mid-Week				
8	Modularization (functions) and introduction of a library - pygame	Lecture (3h), Lab (1.5h).	Class Test / Final Term / Lab work Progressive project	CLO2 CLO3
9	Library - pygame	Lecture (3h), Lab (1.5h).	Class Test / Final Term / Lab work Progressive project	CLO2 CLO3
10	Files (File I/O) - text mode	Lecture (3h), Lab (1.5h).	Class Test / Final Term / Lab work Progressive project	CLO2 CLO3
11	Exception Handling - try-catch	Lecture (3h), Lab (1.5h).	Class Test / Final Term / Lab work Progressive project	CLO3
12	Python - Class, Object	Lecture (3h), Lab (1.5h).	Progressive project	CLO3
13	REVIEW	Lecture (3h), Lab (1.5h).	Class Test / Final Term / Lab work Progressive project	CLO3
14	REVIEW	Lecture (3h), Lab (1.5h).	Class Test / Final Term / Lab work Progressive project	CLO3

ASSESSMENT AND EVALUATION

Assessment Type	Assessment Tools	Marks Distribution	Assessment No	CLO Assessed	Blooms Category	Sub Total
Continuous Assessment	Class Tests	15%	Mid Test - 1			50%
			Mid Test -2			
			Final Test - 3			
			Final Test - 4			
	Lab Assessment	10%		CLO3	Determine, Apply, Develop	
	Progressive Project	15%				
	Tutorial	10%				
Summative Assessment	Mid Term Examination	20%		CLO1	Determine, Apply	50%
	Final Term Examination	30%		CLO2	Analyze, Determine, Apply, Develop	

Total	**100%**

CLO 1 Evaluation Rubric		**Blooms Level: Analyze**
Evaluation Category	**Evaluation Definition**	**Marks Allocation Max – 25 CLO pass – 60%**
Program Compile & Run	Build and run program using IDE, show desired output in the console window	Max – 5 | Range 0 - 5
Logical Operations	Apply logical operations to connect two or more expressions. Use logical operator such as AND, OR, NOT and so on.	Max – 5 | Range 0 - 5
Correctness	Make the program error free, mathematically, and logically make it accurate	Max – 5 | Range 0 - 5
Completeness	Complete the project with implementing all the given requirements	Max – 5 | Range 0 - 5
Coding Standard	Improve readability and maintainability of the code, write code in a way to make it easy for reuse, to figure out mistakes	Max – 5 | Range 0 - 5

CLO 2 Evaluation Rubric		**Blooms Level: Determine, Apply**
Evaluation Category	**Evaluation Definition**	**Marks Allocation Max – 25 CLO pass – 60%**
Solution Design	Conceptual design of the proposed solution.	Max – 5 | Range 0 - 5
Design Correctness	Explain relevance of the solution in relation to the stated problem.	Max – 5 | Range 0 - 5
Implementation	Correctness of the developed program – output, coding standard and others.	Max – 5 | Range 0 - 5
Completeness	Comprehensiveness of the program in relation to problem definition – does it meets all the requirements?	Max – 5 | Range 0 - 5
Explain	Able to defend / reason on the solution they did in relation to the problem definition, e.g., why a specific approach is used not the other?	Max – 5 | Range 0 - 5

CLO 3 Evaluation Rubric		**Blooms Level: Develop**
Evaluation Category	**Evaluation Definition**	**Marks Allocation Max – 25 CLO pass – 60%**
Solution Design	Conceptual design of the proposed solution.	Max – 5 | Range 0 - 5
Design Correctness	Explain relevance of the solution in relation to the stated problem.	Max – 5 | Range 0 - 5
Implementation	Correctness of the developed program – output, coding standard and others.	Max – 5 | Range 0 - 5

Completeness	Comprehensiveness of the program in relation to problem definition – does it meets all the requirements?	Max – 5 | Range 0 - 5
Explain	Able to defend / reason on the solution they did in relation to the problem definition, e.g., why a specific approach is used not the other?	Max – 5 | Range 0 - 5

REFERENCE BOOK AND ADDITIONAL MATERIALS

a. Python Crash Course, 2nd Edition: A Hands-On, Project-Based Introduction to Programming, Eric Matthes

b. Learning Python, 5th Edition, Mark Lutz

c. Automate the Boring Stuff with Python, 2nd Edition: Practical Programming for Total Beginners, Al Sweigart

VERIFICATION

Prepared by:

Course Coordinator
Date:

Moderated by:

Lead
OBE Committee for CSE
Date:

Checked by:

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Head, Dept of CSE, IUB
Date:

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